Practice workbook · Reading a sine model: values, biggest and smallest

Reading off values of a sinusoidal model or graph by substituting and using exact values and the range of sine and cosine.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-01

1. Question 1 · part b.ii · 2 marks

2020 NOV · Paper 01 (regional R)

Worked-example retry · Attempt part b.ii only.

DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

1 Here is a formula

$$P = 3 + 2\sin\left(\frac{3\pi t}{8}\right) \qquad 0 \leqslant t \leqslant 12$$

(a) Find the exact value of P when $t = \frac{10}{3}$

(2)

(b) Find

(i) the largest value of P

(ii) the smallest value of P

(2)

(c) Find the least value of t for which $P = 4$

(3)

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1. Question 1 · part b.ii · 2 marks · Working and answer

Working and answer

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2. Question 1 · part a · 2 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part a only.

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2. Question 1 · part a · 2 marks · Working and answer

Working and answer

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3. Question 1 · part b.i · 0 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b.i only.

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1 Here is a formula

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(3)

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3. Question 1 · part b.i · 0 marks · Working and answer

Working and answer

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4. Question 12 · part a · 1 mark

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Further practice · Attempt part a only.

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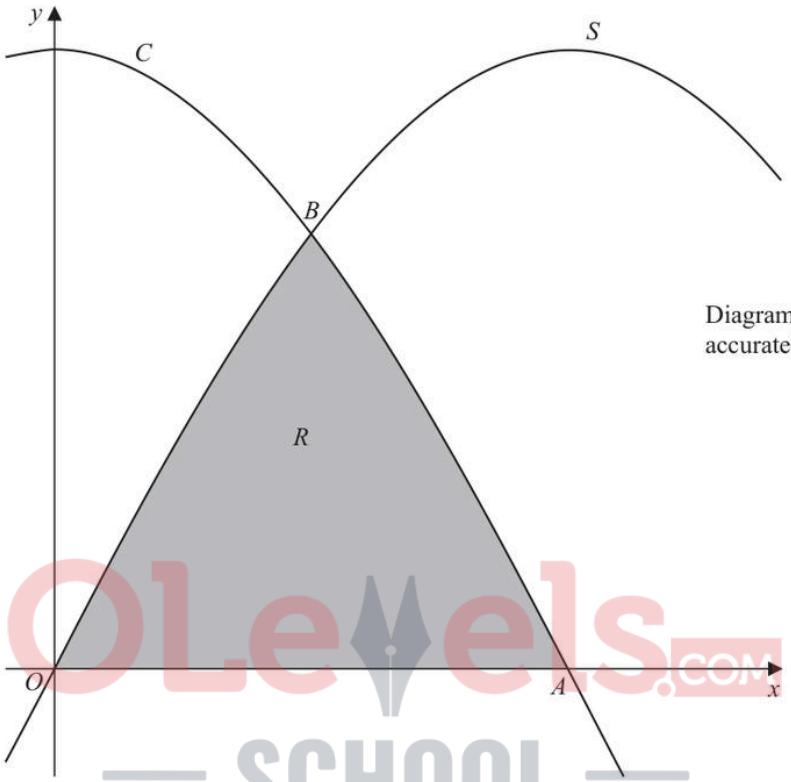


Diagram **NOT** accurately drawn

Figure 2

The region R, shown shaded in Figure 2, is bounded by the x-axis, the curve S with equation $y = 2\sin x$ and the curve C with equation $y = 2\cos x$.
As shown in Figure 2, C crosses the x-axis at the point A.

(a) Write down the x coordinate of A.

(1)

As shown in Figure 2, C and S intersect at the point B.

(b) Find the x coordinate of B.

(2)

(c) Using calculus, find the area of the shaded region R.
Give your answer in the form $a - \sqrt{b}$ where a and b are integers.

(4)

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P 6 2 2 8 3 A 0 3 8 4 0

4. Question 12 · part a · 1 mark · Working and answer

Working and answer

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